

Number theory – 7

The integer part, sizes of roots and denominators

Homework

24 сентября 2026 года

Theoretical minimum. One has $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$, $\lfloor \lfloor x \rfloor / n \rfloor = \lfloor x / n \rfloor$ and Hermite's identity $\sum_{k=0}^{n-1} \lfloor x + \frac{k}{n} \rfloor = \lfloor nx \rfloor$. A sum $\sum \lfloor f(k) \rfloor$ counts lattice points under a graph; for coprime p, q this gives $\sum_{k=1}^{q-1} \lfloor \frac{kp}{q} \rfloor = \frac{(p-1)(q-1)}{2}$. Rayleigh–Beatty: for irrational $\alpha, \beta > 1$ with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$ the sequences $\lfloor n\alpha \rfloor$ and $\lfloor n\beta \rfloor$ partition the positive integers. Roots are located by $1 + \frac{t}{a} - \frac{(a-1)t^2}{2a^2} \leq (1+t)^{1/a} \leq 1 + \frac{t}{a}$ ($t \geq 0$) and exact comparison with integer powers. For $L(n) = \text{lcm}(1, \dots, n)$ one has $L(N) \int_0^1 P \in \mathbb{Z}$ for $P \in \mathbb{Z}[x]$ of degree $< N$, whence $2^n \leq L(n) \leq 4^n$ for $n \geq 7$. A sum of rationals in which exactly one term has the least p -adic valuation has that valuation. Landau's function $g(n) = \max_{\sigma \in S_n} \text{ord } \sigma$ satisfies $\ln g(n) \sim \sqrt{n \ln n}$.

1. Prove the converse of the Rayleigh–Beatty theorem: if $\alpha, \beta > 0$ and every positive integer occurs exactly once among the numbers $\lfloor n\alpha \rfloor, \lfloor n\beta \rfloor$ ($n \geq 1$), then α, β are irrational and $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. Then prove that the n -th positive integer which is not a perfect square equals $n + \lfloor \sqrt{n} + \frac{1}{2} \rfloor$, and describe the positive integers not of the form $\lfloor k\sqrt{2} \rfloor$. (For α the golden ratio see Game theory – 1.)
2. Prove that for every positive integer n $\lfloor \sqrt{n} + \sqrt{n+1} \rfloor = \lfloor \sqrt{4n+2} \rfloor$ and $\lfloor \sqrt[3]{n} + \sqrt[3]{n+1} + \sqrt[3]{n+2} \rfloor = \lfloor \sqrt[3]{27n+26} \rfloor$.
3. What is the units (i.e., rightmost) digit of $\left\lfloor \frac{10^{20000}}{10^{100} + 3} \right\rfloor$?
(Putnam 1986 A2)
4. Prove that $\frac{1}{m} + \frac{1}{m+1} + \dots + \frac{1}{n}$ is not an integer for $1 \leq m < n$. Prove that the reduced denominator of $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ divides $L(n)$ and

is divisible by $2^{\lfloor \log_2 n \rfloor}$ and by every prime p with $\frac{n}{2} < p \leq n$, and show that it can be strictly smaller than $L(n)$.

5. Let n be a positive integer and let $a_1 \leq a_2 \leq \dots \leq a_n$ be real numbers such that $a_1 + 2a_2 + \dots + na_n = 0$. Prove that $a_1 \lfloor x \rfloor + a_2 \lfloor 2x \rfloor + \dots + a_n \lfloor nx \rfloor \geq 0$ for every real number x . (VJIMC 2018)

6. Find all real numbers $x > 1$ such that for every positive integer n the equality

$$\left\lfloor \frac{n+1}{x} \right\rfloor = n - \left\lfloor \frac{n}{x} \right\rfloor + \left\lfloor \frac{\lfloor n/x \rfloor}{x} \right\rfloor - \left\lfloor \frac{\lfloor \lfloor n/x \rfloor / x \rfloor}{x} \right\rfloor + \dots$$

holds (the sum on the right is finite). (VJIMC 2025)

7. Prove that Landau's function $g(n)$ equals the maximum of $q_1 q_2 \dots q_r$ over all collections of powers $q_1, \dots, q_r$ of distinct primes with $q_1 + \dots + q_r \leq n$, and that there are absolute constants $c_1, c_2 > 0$ with $c_1 \sqrt{n \ln n} \leq \ln g(n) \leq c_2 \sqrt{n \ln n}$ for $n \geq 2$. Use only Chebyshev's bounds for $\pi(x)$; the case of small n is in Abstract algebra – 3.

8. Let n be a positive integer. Prove that $\sum_{k=1}^n (-1)^{\lfloor k(\sqrt{2}-1) \rfloor} \geq 0$. (Putnam 2020 B6)

9. For a positive real number α define $S(\alpha) = \{\lfloor n\alpha \rfloor : n = 1, 2, 3, \dots\}$. Prove that $\{1, 2, 3, \dots\}$ cannot be expressed as the disjoint union of three sets $S(\alpha)$, $S(\beta)$ and $S(\gamma)$. (Putnam 1995 B6)

10. For every positive integer n , let $f(n), g(n)$ be the minimal positive integers such that $1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} = \frac{f(n)}{g(n)}$. Determine whether there exists a positive integer n for which $g(n) > n^{0.999n}$. (IMC 2023)